

Blood Vessel Segmentation Project

Image Processing Approach

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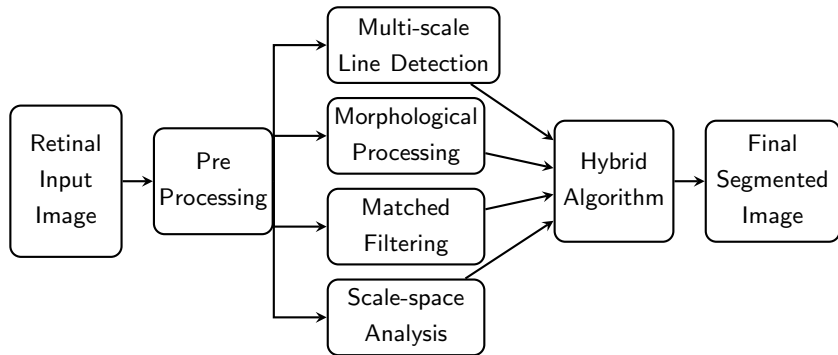
What is Blood Vessel Segmentation?

- Blood vessel segmentation is the process of automatically detecting and extracting retinal blood vessels from fundus images.
- These vessels are crucial biomarkers for diseases such as diabetic retinopathy, glaucoma, and hypertension.
- Segmenting vessels helps doctors analyze vessel width, branching patterns, and leakage regions.
- Manual segmentation is slow, labour-intensive, and requires expert ophthalmologists.
- Automated segmentation improves:
 - Early disease detection
 - Diagnostic accuracy
 - Screening efficiency for large populations
- It also serves as the first step for many downstream tasks (optic disc detection, lesion detection, severity grading).

Aim of the Project

- Automatic blood vessel segmentation in retinal fundus images can greatly support early diagnosis of diabetic retinopathy.
- It helps less-experienced clinicians make more accurate assessments by extracting clear vessel structures.
- Many algorithms exist for vessel segmentation across different imaging systems and disease conditions.
- **Goal of this project:**
 - Evaluate multiple existing vessel segmentation algorithms.
 - Compare their performance using the DRIVE dataset.
 - Combine the strengths of individual methods to achieve improved segmentation accuracy.

System Pipeline



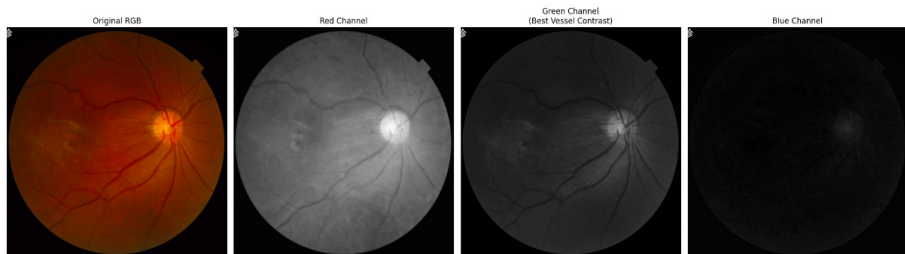
Pre Preprocessing

Why Preprocessing?

- Preprocessing enhances vessel visibility and suppresses noise so that all four segmentation algorithms work more reliably.

Green Channel Extraction

- The green channel provides the **highest contrast** between vessels and background in fundus images.



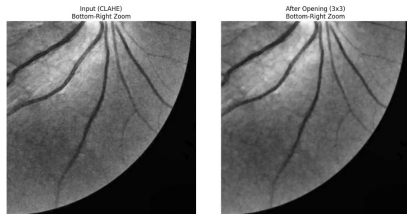
Pre Preprocessing

Histogram Equalization

- Used to correct uneven illumination across the retinal image.
- Enhances local contrast so that vessels become more distinguishable.
- Adaptive histogram equalization (CLAHE) is used for better results.

Morphological Opening

- Removes the central vessel light reflex using a small disk structuring element (3-pixel radius).
- Smooths bright ridges on vessels to avoid false detections.



Morphological Opening Result



Histogram Equalization Result

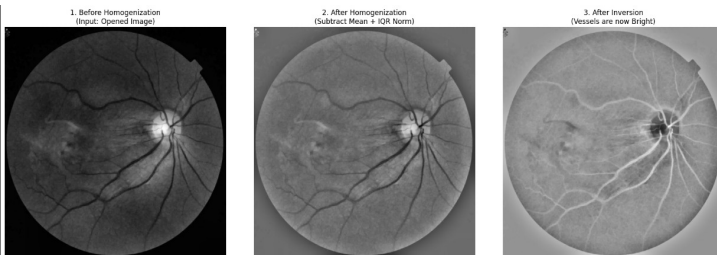
Pre Preprocessing

Background Homogenization

- A large mean filter (69×69) estimates local background.
- Subtracting it equalizes illumination across the image.
- Pixel intensities are then rescaled to 0–1 for uniformity.

Green Channel Inversion

- The green channel is inverted to make vessels appear brighter.
- This improves compatibility with algorithms that detect bright ridges.
- Helps unify vessel appearance across preprocessing steps.



Why Multiple Algorithms for Vessel Segmentation?

- Retinal vessels vary greatly in **size, shape, orientation, and brightness**.
- No single algorithm can perfectly capture:
 - Very thin vessels
 - Thick main vessels
 - Curved structures
 - Low-contrast regions
 - Bright lesions and optic disc interference
- Different algorithms specialize in different strengths:
 - **Matched Filtering** – strong for smooth Gaussian-shaped vessels
 - **Multi-scale Line Detection** – excellent for thin vessels
 - **Scale-space Analysis** – detects vessels across multiple widths
 - **Morphological Processing** – robust around bright lesions and optic disc
- **Therefore:** Running the algorithms in parallel allows each to contribute its strengths.
- The outputs are later combined using a **Hybrid Algorithm** to obtain a more accurate final segmentation.

Matched Filtering – Working

Why this method? Retinal vessels have a **Gaussian intensity profile**, low curvature, and predictable widths (2–10 px).

How it works:

- Extract green channel for highest vessel–background contrast.
- Convolve image with a set of 12 Gaussian kernels at 15° intervals.
- Each filter approximates a linear vessel segment.
- For each pixel, choose the direction giving maximum response.
- Post-processing removes tiny isolated responses.

Output: Continuous probability map of vessel likelihood.

Matched Filtering – Results & Pros/Cons

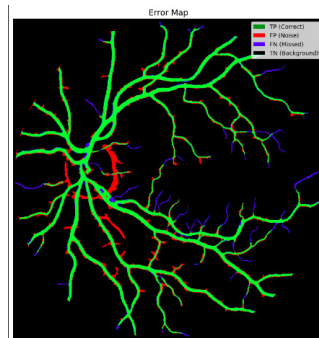
Results: Accuracy 86.88%, Sensitivity 88.55%, Precision 4.44%, F1 - Score 57.55

Advantages:

- Good at detecting medium and large vessels.
- Simple and computationally efficient.

Disadvantages:

- Poor detection near bright regions (optic disc).
- Misses thin / low-contrast vessels.
- High false positives without post-processing.



Sample Matched Filtering Output

Morphological Processing – Working

Goal: Enhance vessel structures and suppress background using multi-scale morphology.

Key Idea:

- Vessels appear as bright, thin ridges in the inverted green channel.
- Morphological top-hat operations can extract dark structures of different widths.

Steps in the Algorithm:

- Invert the green channel image.
- Apply **morphological closing** using a small disk (radius = 2 px) to reduce noise.
- Apply **modified top-hat transform** with disk structuring elements of radius 1–8 px.
- Combine multi-scale outputs using weighted maximum response.
- Normalize output to the range $[0, 1]$.

Morphological Processing – Results & Pros/Cons

Results:

Accuracy $\approx 92.19\%$

Sensitivity $\approx 40.86\%$

Precision $\approx 95.48\%$

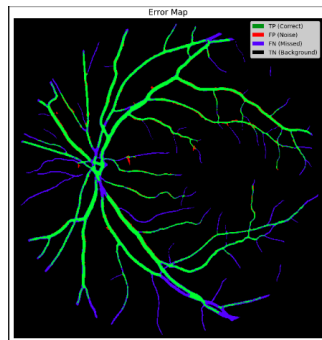
F1-Score $\approx 56.98\%$

Advantages:

- Very high precision (few false positives).
- Performs best near optic disc and bright regions. Effectively enhances vessels of multiple widths.

Disadvantages:

- Low sensitivity — misses many thin vessels.
- Highly dependent on structuring element size.
- Requires multiple scales $\rightarrow$ higher computation



Multi-scale Line Detection – Working

Why this method? Vessels resemble linear structures at multiple scales.
Steps:

- Preprocess: invert green channel + background homogenization.
- Detect lines at 7 scales (lengths: 3, 5, ... 15 pixels).
- For each scale, compute line response across 18 orientations.
- For each pixel, take strongest oriented response.
- Weight large scales more to detect thicker vessels.
- Combine multi-scale responses $\rightarrow$ normalize to $[0,1]$.

Multi-scale Line Detection – Results & Pros/Cons

Results:

Accuracy $\approx 93.21\%$

Sensitivity $\approx 87.69\%$

Precision $\approx 66.59\%$

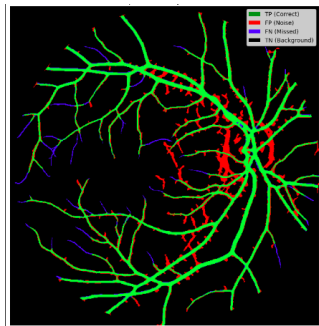
F1-Score $\approx 75.68\%$

Advantages:

- Excellent at capturing thin + thick vessels.
- High sensitivity (detects more small vessels).

Disadvantages:

- More false positives than other methods.
- Sensitive to background noise if preprocessing is weak.



Scale-space Analysis – Working

Why this method? Vessels appear at different thicknesses $\rightarrow$ analyzing across scales captures full structure.

Steps:

- Convolve image with Gaussian kernels at multiple scales (= 1.5 $\rightarrow$ 10).
- Compute gradient magnitude = vessel edges.
- Compute Hessian eigenvalues = ridge strength.
- Weight small scales more ($1/\sigma$).
- Final map = pixelwise max across all scales.

Scale-space Analysis – Results & Pros/Cons

Results:

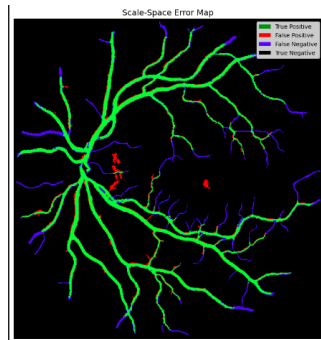
Accuracy $\approx 94.67\%$
Sensitivity $\approx 71.49\%$
Precision $\approx 83.18\%$
F1-Score $\approx 76.88\%$

Advantages:

- Very good precision (few false positives).
- Strong at detecting medium-sized vessels.

Disadvantages:

- Computationally expensive.
- Weak performance on very small vessels.



Hybrid Algorithm – How It Works

Why a Hybrid Approach?

- Each segmentation algorithm has different strengths:
 - Line detection → high sensitivity (captures thin vessels)
 - Morphological processing → excellent near optic disc
 - Matched filtering → good for medium/large vessels
 - Scale-space → strong precision for mid-sized vessels
- Combining them reduces noise and compensates for weaknesses.

How the Hybrid Algorithm Works:

- Each algorithm produces a **vessel confidence map** (values 0–1).
- A pixel's final confidence is the **average of all 4 maps**.
- For bright regions (optic disc/exudates), the **morphological result is used** because it performs best there.

Output: High-quality binary vessel map produced by combining different algorithms.

Hybrid Algorithm – Results

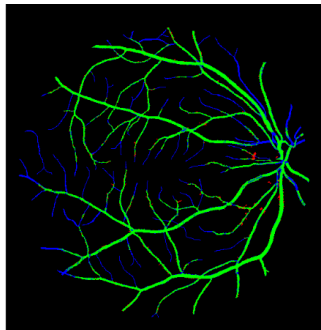
Results:

Accuracy $\approx 94.30\%$

Sensitivity $\approx 76.32\%$

Precision $\approx 83.18\%$

F1 Score ≈ 76.70



Comparison of Algorithms

